

Function Improvements Based on Surface Tags

The clarification about the different surface tags helps tailor the function better. Here's how we'll handle each tag and integrate them into a clean, modular function.

Updated Approach to Tags

1. Quadrilateral Surfaces:

- **Quad Regular:** These are rectangular and follow the fixed dimensions: $\text{distance_ribs} / 2$ and $\text{distance_stringers}$.
- **Quad Irregular:** Quadrilateral surfaces that do not conform to these fixed dimensions.
- **Quad Irregular P1 Inserted and Quad Irregular P4 Inserted:**
 - Subdivisions of **Quad Regular**, split in a non-proportional way. We'll handle these with specific logic for `node_1` and `node_4` assignments.
- **Quad Irregular Root:** Quadrilateral surfaces near the root. These can be highlighted separately for easy identification.

2. Triangular Surfaces:

- **Tri:** These are close to the front spar and will be assigned specific nodes based on their proximity.
- **Tri Root:** Similar to `tri`, but specifically located at the root for root-specific geometries.

3. Pentagonal Surfaces:

- **Penta Root:** These are pentagonal surfaces close to the root. Future pentagonal surfaces near the front spar will follow similar logic but can be handled separately when they arise.